

Exercise 8-1: Analyze the Forest Fires data

Did you use a language model?

A : No, I did not find any use for a language model for this assignment except for writing comments.

Read the data

```
In [1]: import pandas as pd
```

```
In [2]: # Read a pickle file containing data on fires by month
fires_by_month = pd.read_pickle('fires_by_month.pkl')
```

```
In [3]: # Display the first few rows of the fires_by_month DataFrame
fires_by_month.head()
```

```
Out[3]:
```

	state	fire_year	fire_month	acres_burned	days_burning	fire_count
0	AK	1992	5	4202.0	135.0	14
1	AK	1992	6	86401.0	417.0	23
2	AK	1992	7	48516.7	500.0	26
3	AK	1992	8	3305.0	92.0	4
4	AK	1992	9	20.0	1.0	1

Group and aggregate the data

```
In [4]: # Group the data by state and fire_year, and sum the values
fires_grouped = fires_by_month.groupby(['state', 'fire_year'])
```

C:\Users\Dell\AppData\Local\Temp\ipykernel_5456\824044352.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
fires_grouped = fires_by_month.groupby(['state', 'fire_year'])
```

```
In [5]: # Group the data by state and fire_year, and sum the values
fires_by_year = fires_grouped.sum()
```

```
In [6]: # Display the first three rows of the fires_by_year DataFrame
fires_by_year.head(3)
```

```
Out[6]:
```

		fire_month	acres_burned	days_burning	fire_count
state	fire_year				
AK	1992	35	142444.7	1145.0	68
	1993	30	686630.5	3373.0	144
	1994	48	261604.7	2517.0	126

```
In [7]: # Drop the 'fire_month' column from fires_by_year DataFrame
fires_by_year.drop(columns='fire_month', inplace=True)
```

Use pivot tables

```
In [8]: # Filter the data for fire_year greater than or equal to 2013, and reset the index
fires_recent = fires_by_year.query('fire_year >= 2013').reset_index()
```

```
In [9]: # Display the first five rows of the fires_recent DataFrame
fires_recent.head(5)
```

```
Out[9]:
```

	state	fire_year	acres_burned	days_burning	fire_count
0	AK	2013	1320518.50	4644.0	153
1	AK	2014	233426.40	571.0	43
2	AK	2015	5136372.99	16636.0	340
3	AL	2013	28015.65	115.0	581
4	AL	2014	35310.20	107.0	758

```
In [10]: # Pivot the fires_recent DataFrame with index as 'state', columns as 'fire_year'
fires_recent.pivot(index='state', columns='fire_year', values='acres_burned').head(5)
```

```
Out[10]:
```

	fire_year	2013	2014	2015
state				
AK		1320518.50	233426.40	5136372.99
AL		28015.65	35310.20	28386.96
AR		11797.60	18521.80	16302.96
AZ		102307.47	202212.51	154720.38
CA		575555.65	543033.33	844674.02

```
In [11]: # Query fires_by_month for fire_year greater than or equal to 2013, pivot the table
# 'fire_year' as columns, summing 'acres_burned' values, and display the first five rows
fires_by_month.query('fire_year >= 2013').pivot_table(
    index='state', columns='fire_year', values='acres_burned', aggfunc='sum').head(5)
```

Out[11]:

fire_year	2013	2014	2015
-----------	------	------	------

state			
AK	1320518.50	233426.40	5136372.99
AL	28015.65	35310.20	28386.96
AR	11797.60	18521.80	16302.96
AZ	102307.47	202212.51	154720.38
CA	575555.65	543033.33	844674.02

Work with bins

```
In [12]: # Reset the index of fires_by_year DataFrame
fires_by_year.reset_index(inplace=True)
# Display the fires_by_year DataFrame
fires_by_year
```

Out[12]:

	state	fire_year	acres_burned	days_burning	fire_count
0	AK	1992	142444.70	1145.0	68
1	AK	1993	686630.50	3373.0	144
2	AK	1994	261604.70	2517.0	126
3	AK	1995	43762.60	880.0	48
4	AK	1996	598407.20	2112.0	88
...	...	...	...	...	...
1219	WY	2011	134336.25	669.0	160
1220	WY	2012	481206.46	1802.0	215
1221	WY	2013	53884.90	742.0	40
1222	WY	2014	7179.43	224.0	48
1223	WY	2015	36233.50	351.0	107

1224 rows × 5 columns

```
In [13]: # Create a new column 'decade' in fires_by_year DataFrame based on the decade of
fires_by_year['decade'] = pd.cut(fires_by_year.fire_year,
                                bins=[1990,2000,2010,2020],
                                labels=['1990s','2000s','2010s'],
                                right=False)
```

```
In [14]: # Display the first 25 rows of fires_by_year DataFrame
fires_by_year.head(25)
```

Out[14]:

	state	fire_year	acres_burned	days_burning	fire_count	decade
0	AK	1992	1.424447e+05	1145.0	68	1990s
1	AK	1993	6.866305e+05	3373.0	144	1990s
2	AK	1994	2.616047e+05	2517.0	126	1990s
3	AK	1995	4.376260e+04	880.0	48	1990s
4	AK	1996	5.984072e+05	2112.0	88	1990s
5	AK	1997	2.025818e+06	2295.0	128	1990s
6	AK	1998	1.206034e+05	435.0	31	1990s
7	AK	1999	1.005597e+06	2312.0	101	1990s
8	AK	2000	7.561614e+05	1745.0	51	2000s
9	AK	2001	2.197077e+05	1659.0	26	2000s
10	AK	2002	2.186329e+06	3207.0	100	2000s
11	AK	2003	6.013995e+05	1061.0	56	2000s
12	AK	2004	6.591431e+06	7783.0	167	2000s
13	AK	2005	4.635536e+06	7430.0	200	2000s
14	AK	2006	2.543659e+05	699.0	35	2000s
15	AK	2007	6.341360e+05	3506.0	139	2000s
16	AK	2008	9.226770e+04	1695.0	67	2000s
17	AK	2009	2.949289e+06	5389.0	133	2000s
18	AK	2010	1.137138e+06	6223.0	214	2010s
19	AK	2011	2.927164e+05	1686.0	70	2010s
20	AK	2012	3.003448e+05	1265.0	95	2010s
21	AK	2013	1.320519e+06	4644.0	153	2010s
22	AK	2014	2.334264e+05	571.0	43	2010s
23	AK	2015	5.136373e+06	16636.0	340	2010s
24	AL	1992	1.098000e+03	1.0	4	1990s

In [15]:

```
# Drop the 'fire_year' column from fires_by_year DataFrame to prepare for groupby
fires_by_decade = fires_by_year.drop(columns=['fire_year'])

# Display the first few rows of fires_by_decade DataFrame
fires_by_decade.head()
```

```
Out[15]:
```

	state	acres_burned	days_burning	fire_count	decade
0	AK	142444.7	1145.0	68	1990s
1	AK	686630.5	3373.0	144	1990s
2	AK	261604.7	2517.0	126	1990s
3	AK	43762.6	880.0	48	1990s
4	AK	598407.2	2112.0	88	1990s

```
In [16]: # Group the fires_by_decade DataFrame by 'state' and 'decade', and sum the value
fires_by_decade = fires_by_decade.groupby(['state','decade']).sum()
```

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```
fires_by_decade = fires_by_decade.groupby(['state','decade']).sum()
```

```
In [17]: # Display the first few rows of fires_by_decade DataFrame
fires_by_decade.head()
```

```
Out[17]:
```

	state	decade	acres_burned	days_burning	fire_count
AK	1990s	4.884868e+06	15069.0	734	
	2000s	1.892062e+07	34174.0	974	
	2010s	8.420518e+06	31025.0	915	
AL	1990s	1.945928e+05	214.0	216	
	2000s	4.088805e+05	1863.0	535	

```
In [ ]:
```